Lab 08 Contribution Journal

ITAI 1371 | Lab 08: Bias-Variance Tradeoff | Team: TheBetaThetas

Brandy Gonzales

I opened the Lab 8 starter notebook in Jupyter on my laptop. I fixed the code typos and indentation errors, and I replaced the scrambled Task 2 cell so the learning curve graphs displayed properly. I ran each cell and examined the degree 1, 4, and 15 graphs and the learning curves. I answered the Knowledge Check questions, starting with Q1 on which degree underfits, overfits, and fits well. I set up the team's shared Colab notebook so everyone has a spot for their reflection and contribution.

Derrick K. Gibbs-McGlaston

I used Google Colab to run the notebook and review the results for the bias-variance tradeoff activity. The lab compared polynomial regression models with degrees 1, 4, and 15 using sample data based on a sine wave with added noise.

I ran the notebook cells and checked the output graphs. I reviewed the plot comparing the three models and used it to identify degree 1 as underfitting, degree 4 as a better fit for the overall pattern, and degree 15 as overfitting. I also examined the learning curves for degrees 1 and 15, paying attention to how the training and cross-validation scores changed as the number of training examples increased.

I completed the three knowledge-check responses using the graphs produced in the notebook. I explained why the degree 1 model was too simple, why the degree 4 curve followed the data more appropriately, and why the degree 15 model showed problems when evaluated on other data. I also reviewed the unusually large negative cross-validation scores shown for degree 15.

I exported the executed notebook as a PDF with the code, graphs, and answers.

Victor Djuma

My individual contribution to this lab was completing the Bias-Variance Tradeoff notebook in Google Colab, running the required code, reviewing the model outputs, interpreting the graphs, completing the Knowledge Check, and checking the final PDF before submission.

I first worked with the setup section of the notebook, which created a synthetic dataset based on a sine-wave pattern with random noise. I ran the code that generated the data and reviewed the scatter plot so I could understand the pattern the models were supposed to learn. This gave me a starting point for comparing models with different levels of complexity.

For Task 1, I worked with polynomial regression models using degrees 1, 4, and 15. I ran the code that created each model using PolynomialFeatures and LinearRegression. I trained the models on the synthetic dataset, generated predictions, and displayed the three model predictions on the same graph. I then reviewed the graph and compared how each model followed the data.

From this comparison, I identified the degree 1 model as the underfitting model because it was too simple to follow the curved pattern in the data. I identified the degree 15 model as the overfitting model because it was much more complex and followed the training data and noise too closely. I also reviewed the degree 4 model as the better balance between these two extremes because it followed

the main pattern without becoming too complicated.

For Task 2, I worked with the learning_curve function from scikit-learn. I ran the code that created a learning curve for the degree 1 model and another learning curve for the degree 15 model. I reviewed the training and cross-validation scores on the graphs and used them to understand the difference between high bias and high variance. The learning curves helped me see that model performance should be evaluated not only on the data used for training but also on data that was not used to fit the model.

I also completed all three Knowledge Check questions in my own words. For the first question, I explained why the degree 1 model represents underfitting, why the degree 15 model represents overfitting, and why degree 4 provides a better balance. For the second question, I explained what the learning curve for the underfitting model shows and why adding more data does not completely solve a problem caused by a model that is too simple. For the third question, I explained that the gap between the training and cross-validation scores for the complex model is a sign of high variance and poor generalization.

I also worked through technical problems that affected the completion of the assignment. The original notebook had formatting problems where some of the text and code did not display with the correct line breaks, so I used the corrected notebook before continuing. After completing the notebook, I exported it to PDF and reviewed the entire document instead of submitting it immediately. I noticed that the final page had a formatting issue: the second learning-curve graph overlapped with the Knowledge Check, cutting off some of the answers. I adjusted the PDF export and checked the new version again. The final PDF displayed both learning curves and all three answers correctly.

My contribution was therefore not only running the code. I also reviewed the outputs, interpreted what the different models were showing, completed the written analysis, checked the notebook for correct execution, and verified that the final PDF was complete and readable. These were the specific parts of the lab that I personally worked through.

Completing these steps helped me contribute to the project while also improving my own understanding of the concepts. I now have a better understanding of how model complexity can lead to underfitting or overfitting and how learning curves can help identify these problems.